

TECHNICAL SKILLS & CORE COMPETENCIES		CORE // STACK
Languages & Core:	Python, TypeScript, JavaScript (ES6+), SQL, C/C++, PHP, HTML5, CSS3 / Tailwind CSS	
AI & Machine Learning:	PyTorch, TensorFlow, Scikit-Learn, LangChain, Hugging Face, LLMs, Autonomous Agents, Agent-Based Modeling, Reinforcement Learning, Multi-Modal Gemini API, Model Fine-Tuning, Vector Embeddings, Hybrid RAG, NLP	
Computer Vision & Audio:	OpenCV, Computer Vision, Web Audio API (Audio Graph Rendering), Canvas 2D (60 FPS), Edge AI, Timeline Compositing, Data Visualization (Matplotlib, Seaborn, Pandas, NumPy)	
Full-Stack & Frameworks:	React 19 / 18, Next.js, React Native, Node.js, NestJS, FastAPI, Laravel, REST APIs, WebSockets, Tkinter, WordPress (CMS & Headless), Zustand, AsyncIO	
Databases & Infrastructure:	PostgreSQL (pgvector), Microsoft SQL Server, Redis, PhpMyAdmin, RDBMS Design, Docker, Git, GitHub Actions (CI/CD), Linux/Bash, Mapbox API, Vite	
Methodologies & Practices:	Test-Driven Development (TDD), Agile Methodologies, Software Architecture, Debugging & Profiling, Data Cleaning, Analytical Problem Solving	

SOFTWARE ENGINEERING & AI SYSTEMS EXPERIENCE		SYSTEMS ARCHITECTURE
Klyptic — Browser-Native Non-Linear Video Editor & AI Subtitle Studio Lead Architect		2025 – Present
React 18, TypeScript, HTML5 Canvas 2D, Web Audio API, Tailwind CSS, Zustand		
- Architected an in-browser non-linear editing suite delivering zero-server video rendering and sustained 60 FPS compositing via hardware-accelerated Canvas 2D and requestAnimationFrame loops. - Engineered non-destructive multi-track sequencing with sub-frame keyframe interpolation, procedural transition shaders, and audio-video timeline synchronization. - Integrated local AI auto-captioning engine with millisecond word-timestamp alignment and offline audio graph rendering via Web Audio API context, reducing backend compute costs to zero.		
Trade Bot Gemini — Autonomous Multi-Modal Trading Agent AI Engineer & Creator		2025 – Present
Python, Google Gemini Multi-Modal API, NumPy, Pandas, AsyncIO, Autonomous Reasoning Loops		
- Engineered an autonomous multi-modal quantitative trading agent ingesting live price action, order-book depth, and macroeconomic sentiment to generate structured, risk-weighted hypotheses. - Designed a structured self-auditing reflection loop enforcing strict downside limits, executing decision inference cycles in under 800ms with zero policy breaches. - Implemented dynamic volatility calibration and adaptive position sizing based on real-time rolling drawdown thresholds.		
Pakistan Super App — Mobility & Roadside Logistics Platform Full-Stack Systems Engineer		2024 – 2025
TypeScript, React Native, Node.js, WebSockets, Mapbox API, Redis, Monorepo		
- Architected a cross-platform mobility monorepo unifying passenger, driver, and emergency roadside recovery mobile clients with a scalable Node.js backend service layer. - Implemented a sub-second reverse-auction bidding protocol over persistent bidirectional WebSockets with automated geospatial geohash proximity dispatch. - Engineered battery-optimized background GPS coordinate synchronization with automatic connection failure recovery.		
Movie Recommender AI — Vector Similarity & Hybrid ML Engine ML Developer		2024
Python, Scikit-Learn, Pandas, NLP Embeddings, Truncated SVD, Cosine Similarity		
- Formulated a hybrid recommendation engine synthesizing high-dimensional NLP plot embeddings with sparse collaborative rating matrices to mitigate user cold-start bias. - Optimized vector space retrieval using Truncated SVD dimensionality reduction, achieving < 10ms retrieval latency across large candidate catalogs.		
Full-Stack & AI Systems Consultancy Independent Software Developer (Upwork & Global Clients)		2024 – Present
React, Next.js, Python, PHP, Laravel, WordPress, Docker, PostgreSQL, REST APIs		
- Delivered high-performance web applications, custom CMS solutions (WordPress, PHP, Laravel), and headless modern frontends (Next.js/React) for international clients. - Implemented custom automated data processing pipelines and robust API endpoints with automated unit testing and containerized Docker deployments.		

COMPETITIVE PROGRAMMING & TECHNICAL HONORS		AWARDS & CONTESTS
PROCOM — National SQL & Database Optimization Contest FAST-NUCES		FAST-NUCES
Competed in intensive database query-solving; authored optimized multi-table joins, subqueries, and analytical SQL scripts under strict timed limits.		
SuperComp — Speed Programming & Algorithmic Problem Solving Bahria University (BUKC)		BUKC
Solved complex algorithmic challenges and edge-case test suites in C++ and Python under strict timed conditions, optimizing time and space complexity.		
SuperComp — Code in the Dark (Blind Frontend Engineering Contest) Bahria University (BUKC)		BUKC
Competed in high-pressure blind frontend contest: analyzed an intricate target web interface shown briefly, then reproduced the pixel-precise responsive layout from memory using pure HTML/CSS without live browser previews.		
TEKNOFEST International Aerospace & Technology Festival Karachi Expo Center		Expo Center
Participated in international technology competition tracks, collaborating on technical problem-solving and innovation-focused engineering solutions.		

EDUCATION & FORMAL FOUNDATIONS		ACADEMIC FOUNDATION
Bahria University Karachi — Bachelor of Science in Artificial Intelligence (BS AI)		2023 – Present
Key Coursework: Deep Learning, Neural Networks, Natural Language Processing, Computer Vision, Probabilistic Graphical Models, Data Structures & Algorithms, Database Systems (RDBMS), Systems Architecture.		